Predicting Metabolic Cost During Human-in-the-Loop Exoskeleton Optimisation

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Code: github.com/Mukilskanda/Predicting-Metabolic-Cost-During-Human-in-the-Loop-Optimisation

1. Problem statement

Ankle exoskeletons are personalised by **human-in-the-loop optimisation (HILO)**: a CMA-ES optimiser applies a new torque profile every 2 minutes and keeps the profiles that minimise the wearer's **metabolic cost**. Metabolic cost, however, is measured by indirect calorimetry through a face mask: it needs ~1 min to settle, is very noisy, and only works in the lab. **Goal:** learn a regression model that predicts metabolic cost from signals a wearable could record (surface EMG, gait features, controller settings), so that HILO could run faster and outside the laboratory. We reproduce and extend Krimsky & Ng (Stanford CS229, 2018).

2. Dataset

Stanford Biomechatronics Lab HILO study with bilateral ankle exoskeletons and naive users. **2 subjects × 5 days × 36 control conditions**; one row covers the last 30 s of a 2-min condition. One physically impossible (negative) target is removed, leaving **180 + 179 samples** with **29 features**:

- **9 step/gait**: L/R peak vertical ground-reaction force (÷ body weight), L/R peak dorsi- and plantar-flexion angles, L/R step time and step width, each the per-step median.
- **16 EMG** (8 muscles per leg): band-pass, rectify, low-pass and RMS, normalised by the normal-walking peak of the same day.
- **4 control**: peak torque, peak time, rise time and fall time of the exoskeleton torque profile.
- **Target**: metabolic rate minus the day's standing baseline, ÷ the day's normal-walking rate (1.0 = unassisted walking). Variance: 0.0665 (S1) and 0.0358 (S2). Measurement noise sets an MSE floor of about 0.0025.

The data are tiny (p ≈ n/6), noisy and **non-stationary**: cost falls by ~40 % over the five days as subjects learn to walk with the device (Fig. 1).

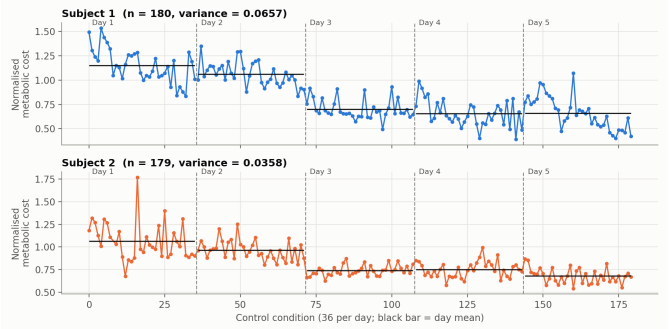


Fig. 1. Measured metabolic cost; black bars mark daily means. Both subjects adapt over the days.

3. Approach

- **Leakage-free, nested evaluation.** Every model is a scikit-learn pipeline, so scaling and all hyper-parameter searches (LASSO λ, SVR C/ε, PLS components, number of NN neurons) run on training folds only. The paper selected λ on the same folds it reported.
- **Two protocols.** (a) Repeated nested 5-fold CV (5 × 5) for comparison. The paper used its own 10-fold CV, so its scores are reference benchmarks, not a like-for-like match. (b) **Leave-one-day-out**: predict an entire unseen session. This is the realistic case, which the paper named as the ideal test but did not run.
- **13 models + ensemble.** OLS, LASSO, Ridge, Elastic Net, PLS, kNN, SVR-RBF, Random Forest, Extra Trees, Gradient Boosting, a **Gaussian Process** (linear + RBF + white-noise kernel, giving calibrated uncertainty) and a **Bayesian-regularised NN**. The NN is our NumPy re-implementation of MATLAB *trainbr*: Levenberg-Marquardt with MacKay evidence updates of α, β and γ.
- **Analyses.** Feature-group ablation, neuron sweep, permutation and LASSO-stability importance, PCA vs. PLS vs. forward selection, learning curves, GP calibration, cross-subject transfer, and *within-day rank correlation* (what the optimiser actually needs).

4. Implementation

Python 3 · NumPy · scikit-learn · Matplotlib. Two layers: `paper_reproduction/`, a 1:1 port of the paper's MATLAB scripts that approximately reproduces its linear-regression baseline (S2 0.0172 vs 0.0168; S1 0.0107 vs 0.0089, as fold splits differ) and is the demo baseline; and `metacost/` + `scripts/` (12 experiments, 16 figures, 10 unit tests incl. a Jacobian check), and a command-line predictor used for the live demo, which trains on 4 days and predicts the unseen 5th.

5. Results

Table 1. Cross-validated MSE, all 29 features, selected models (bold = best of all 13 models per column; same run as the slides).

Model	S1 random	S1 unseen day	S2 random	S2 unseen day
Constant (mean)	0.0665	0.0917	0.0365	0.0482
Linear (OLS)	0.0097	**0.0140**	0.0162	0.0435
LASSO	0.0093	0.0234	0.0137	0.0280
Bayesian-reg. NN	0.0119	0.0214	0.0127	0.0496
PLS	0.0095	0.0145	0.0160	0.0327
SVR (RBF)	0.0087	0.0244	**0.0109**	0.0455
Extra Trees	0.0104	0.0369	0.0109	0.0305
Gradient Boosting	0.0119	0.0426	0.0114	**0.0263**
Gaussian Process	0.0081	0.0175	0.0129	0.0460
Ensemble (avg)	**0.0079**	0.0233	0.0110	0.0322
Paper (Krimsky & Ng)	0.0089	-	0.0168	-

Lower than the paper's reference scores. Under our setup, our best MSE is **0.0079** for S1 (ensemble; R^2 = 0.88, ≈ 9 % error) and **0.0109** for S2 (SVR). The paper reports 0.0089 (S1, linear regression, all features) and 0.0168 (S2, linear regression, step + EMG) under its own CV, so this is not a like-for-like comparison. The Gaussian Process is the most consistent single model.

Unseen days are harder. Leave-one-day-out roughly doubles the error. S1 still generalises (OLS/PLS/Ridge MSE ≈ 0.014, R^2 = 0.79; GP r = 0.88 in Fig. 2) with a within-day Spearman ρ ≈ 0.6. S2 drops to R^2 = 0.27: its day-level drop in cost on days 3-5 is not visible in the sensors. Simple, heavily regularised linear models generalise best across days, while flexible trees and kNN overfit within-day structure.

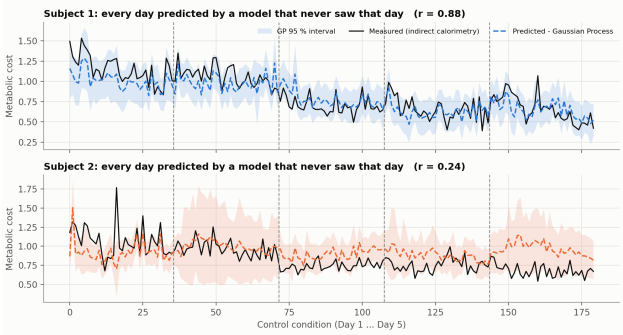


Fig. 2. Each day predicted by a GP trained only on the other four days, shaded by its 95 % interval.

Which signals matter? Table 2 reproduces the paper's ablation: **EMG beats step data** for S1 and for LASSO on S2 (for the GP on S2 the two groups are similar), and dropping the control parameters costs little (S1: 0.0081 → 0.0090). Grouped permutation importance ranks the groups EMG > Step > Control for S1 and EMG > Control > Step for S2. Right-leg muscles R4 and R7 (S1) and R2 (S2) are selected in 93-100 % of 200 LASSO bootstraps, so a reduced sensor set is feasible.

Table 2. Feature-set ablation, MSE (random CV).

Feature set	S1 LASSO	S1 GP	S2 LASSO	S2 GP
All features	0.0093	0.0081	0.0137	0.0129
Step + EMG	0.0099	0.0090	0.0147	0.0120
EMG only	0.0165	0.0107	0.0178	0.0146
Step only	0.0214	0.0151	0.0208	0.0141

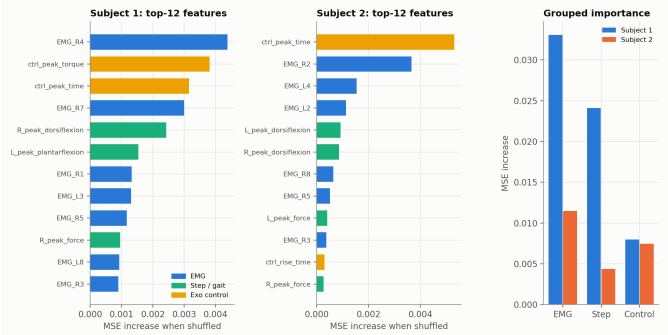


Fig. 3. GP permutation importance (held-out folds): top-12 features per subject and grouped importance.

Dimensionality reduction. PCA ignores the target, which is why the paper found it worked poorly. Supervised **PLS with a single component** (0.0138) matches the best PCA model, which needs 8 components, on S2. **Neural networks:** the evidence framework selects only 1-2 hidden neurons, and the NN never beats the GP. **Across people:** a model trained on S1 and tested on S2 does worse than predicting the mean (MSE 0.152 vs 0.036). Per-subject z-scoring cuts this to 0.046, which is still not useful. **Uncertainty:** the GP's 95 % intervals cover 94 % of unseen-day points for S1 but only 77 % for S2, which correctly flags the harder subject. Subject 1's learning curve (0.0155 → 0.0088) is still falling.

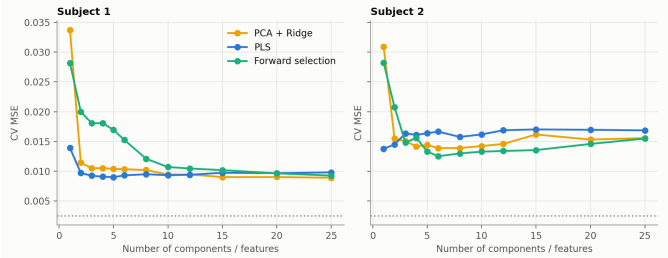


Fig. 4. CV MSE vs. number of components/features. Supervised PLS needs far fewer dimensions than PCA.

6. Reducing unseen-day error with 3 mask spot-checks

Session calibration. If k conditions of a new day are also measured with the mask, the day's predictions can be shifted by the mean residual. A raw mean of k noisy residuals is itself noisy, so we use an **empirical-Bayes shrunken offset**, w·mean(residual) with w = k/(k + σ²/τ²). Here σ² (within-day noise) and τ² (true day-offset variance) are estimated from the training days only. With **3 random spot-checks out of 36** (92 % fewer mask readings), unseen-day MSE for S2 drops **0.0460 → 0.0223 (GP, –51 %)** and 0.0362 → 0.0182 (Ridge). S1 drops 0.0175 → 0.0142 (Fig. 5). Calibrating on the *first* conditions of a session is worse: cost runs high early in each session, so it hurt S1 at k ≥ 5 (Ridge 0.0120 → 0.0143, or 0.0184 without shrinkage).

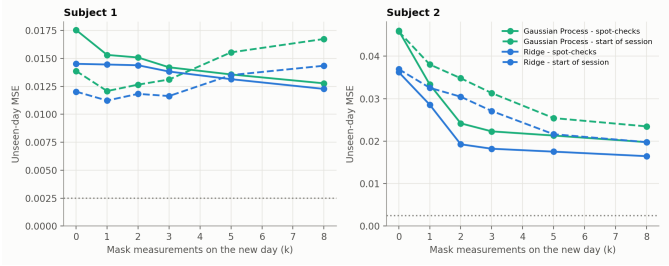


Fig. 5. Unseen-day MSE vs. number of mask measurements on the new day. Solid: random spot-checks; dashed: start of session (scored on conditions 9-36).

Fewer sensors. Keeping only the k EMG channels that enter the LASSO path first (chosen inside each training fold) shows that **4 electrodes beat all 16 for S2**: unseen-day Ridge 0.0362 → 0.0242, and random-CV GP 0.0132 → 0.0114. S1 needs about 8 electrodes to stay within 3 % (Fig. 6). **Transfer:** recalibrating a model from the other person does not help. A model trained on just 5-10 of the new person's own conditions wins.

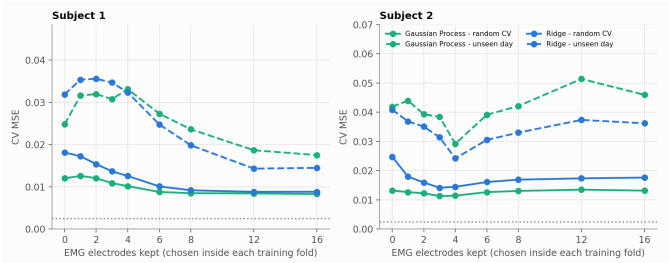


Fig. 6. CV MSE vs. number of EMG electrodes kept (all step and control features always kept). Solid: random CV; dashed: unseen day.

7. Conclusions

- EMG and gait signals predict normalised metabolic cost to within about **9-10 %** within a session (mean baseline: 19-26 %), with lower MSE than the original study's reference scores under our setup.
- Honest evaluation exposed the real weakness: on a completely unseen day the error rises to 12 % (S1) and 16 % (S2), because random splits leak day-level information.
- **Promising results on this small dataset reduce that weakness with very few extra measurements.** With **empirical-Bayes shrinkage**, just **3 mask spot-checks** per 36-condition session (92 % fewer mask readings) cut Subject 2's unseen-day error by **51 %** (GP MSE 0.046 → 0.022) without harming Subject 1. Keeping only **4 of the 16 EMG electrodes** also lowered Subject 2's unseen-day Ridge MSE from 0.036 to **0.024**, pointing to cheaper wearable hardware.
- Models must still be **subject-specific**: cross-person transfer fails even after calibration. These are not proof of deployment readiness. **Future work**: validate on more subjects and real sessions, anthropometrics and IMU kinematics, and the calibrated surrogate as a prior inside CMA-ES or Bayesian optimisation.

References. [1] E. Krimsky, E. Ng, CS229 Stanford, 2018. [2] J. Zhang et al., *Science* 356:1280, 2017. [3] D. MacKay, *Neural Comp.* 4:415, 1992. [4] F. Foresee, M. Hagan, ICNN 1997. [5] C. Rasmussen, C. Williams, *GPML*, MIT Press 2006.